Internal splitting

$$\hat{P}\hat{\mu}\hat{P}$$

At exact degeneracy

$$\kappa = (d_{\max} - d_{\min})/2$$

Nonzero width: linear cusp

Subspace deformation

$$\hat{X} = \hat{Q}(\partial_F \hat{P})\hat{P}$$

For an isolated cluster

$$\hat{H}_Q\hat{X} - \hat{X}\hat{H}_P = \hat{B}$$

Metric: $g = \mathrm{Tr}(\hat{X}^\dagger \hat{X})$

Numerical continuity

$$S_{0F} = \Psi_0^\dagger \Psi_F$$

For the computed branch

Singular values
+ field refinement

Separate model and solver effects